

Care Guidance Summary for PT-4AE48A

Overview

PT-4AE48A is a 24-year-old male with medium model-predicted diabetes risk (49%). Current profile: HbA1c 5.2%, fasting glucose 90 mg/dL, BMI 19. Key protective factors include HbA1c 5.2% is currently in a reassuring range. and Fasting glucose 90 mg/dL does not currently suggest fasting dysglycemia. Main issues to address now: Random glucose 130 mg/dL is mildly elevated and worth monitoring over time. Blood pressure 120/88 mmHg is above the preferred range and should be rechecked.

Risk Summary

medium diabetes risk (49%) based on the latest available model-supported assessment.

Risk Narrative

Risk is in the moderate range. Top features show which inputs the model relied on.

Clinical Snapshot

- Age: 24
- Sex: Male
- Height: 175 cm
- Weight: 68 kg
- HbA1c: 5.2%
- Fasting glucose: 90 mg/dL
- Random glucose: 130 mg/dL
- Total cholesterol: 150 mg/dL
- HDL: 40 mg/dL
- LDL: 90 mg/dL
- BMI: 19
- Blood pressure: 120/88 mmHg
- Smoking: Never
- Drinking: Never
- Exercise: Heavy

Protective Factors

- HbA1c 5.2% is currently in a reassuring range.
- Fasting glucose 90 mg/dL does not currently suggest fasting dysglycemia.
- BMI 19 is within a healthy range for metabolic risk reduction.
- Current heavy exercise pattern is a strong protective lifestyle factor.
- No active smoking exposure is documented.
- Alcohol intake is not adding a current metabolic burden in the profile.

Active Concerns

- Random glucose 130 mg/dL is mildly elevated and worth monitoring over time.
- Blood pressure 120/88 mmHg is above the preferred range and should be rechecked.

Care Guidance Summary for PT-4AE48A

Key Risk Drivers

- BMI (impact +0.63)
- AGE (impact +0.56)
- LDL (impact +0.42)
- SEX (impact +0.42)
- TOTAL CHOLESTEROL (impact +0.40)
- RANDOM GLUCOSE (impact +0.39)

Lifestyle Recommendations

- Maintain the current glucose-friendly eating pattern to preserve HbA1c 5.2% and fasting glucose 90 mg/dL. Focus on eating a balanced diet with plenty of vegetables, whole grains, lean proteins, and healthy fats to help maintain stable blood sugar levels.
- Preserve the current heavy exercise routine while building in recovery, hydration, and blood-pressure awareness. Start with small, manageable goals, such as a 10-minute walk after meals, and gradually increase the duration and intensity of your exercise.
- Because blood pressure is 120/88 mmHg, emphasize sodium awareness, regular aerobic activity, and repeat BP checks. Monitor your blood pressure regularly, as your diastolic blood pressure is slightly elevated. Lifestyle changes like reducing sodium intake, exercising, and managing stress can help.
- Aim for 7-9 hours of quality sleep per night, as poor sleep can affect blood sugar control and overall health.
- Practice stress-reduction techniques like deep breathing, meditation, or yoga to help manage stress levels, which can impact blood sugar control.
- Track meal patterns around higher glucose readings and reinforce carbohydrate quality. Monitor carbohydrate intake by choosing complex carbs (like whole grains, legumes, and vegetables) over simple carbs (like sugary snacks and refined grains).

Medication Considerations

- Primary option for clinician review: Vitamin D (Cholecalciferol). Safe for general metabolic and bone health support; no known contraindications or interactions in this context. Low risk of adverse effects and may provide secondary benefits for metabolic health.
- Alternatives noted for review: Omega-3 Fatty Acids.
- A 24-year-old male with a normal BMI and currently non-diabetic glycemic metrics (HbA1c 5.2%, fasting glucose 90 mg/dL). Diabetes type is undetermined, and the patient has no known comorbidities or medications, with a low diabetes risk score.
- Safety and validation notes: Renal function assessment is recommended before considering metformin or other pharmacologic agents in the future. No pharmacologic intervention is urgently indicated given the patient's normal glycemic metrics and low diabetes risk score. Lifestyle interventions (e.g.,

Care Guidance Summary for PT-4AE48A

nutrition, physical activity) should be prioritized over pharmacologic options at this time.

Monitoring Plan

- Repeat HbA1c at the usual clinician-directed interval to confirm stability from the current 5.2% baseline.
- Monitor fasting or home glucose trends against the current 90 mg/dL baseline.
- Repeat blood pressure readings and compare against the current 120/88 mmHg baseline.
- Document renal function before considering future pharmacologic escalation.
- Review the lipid profile at follow-up so cardiovascular prevention advice stays targeted.

Doctor Considerations

- Creatinine is not documented in the current profile, which limits medication escalation planning.
- Triglycerides are not documented in the current profile.
- Medication safety notes requiring clinician attention: Renal function assessment is recommended before considering metformin or other pharmacologic agents in the future. No pharmacologic intervention is urgently indicated given the patient's normal glycemic metrics and low diabetes risk score. Lifestyle interventions (e.g., nutrition, physical activity) should be prioritized over pharmacologic options at this time.
- No chronic conditions are documented in the current patient profile, so recommendations are driven mainly by vitals, labs, and lifestyle history.

Evidence Summary

- Standards of Care in Diabetes
- ADA Clinical Guidelines - The Standards of Care in Diabetes (Standards of Care) includes all current clinical practice recommendations of the American Diabetes Association (ADA) and ... - <https://professional.diabetes.org/standards-of-care>
- The American Diabetes Association Releases the Standards of Care ... - Today, the Standards of Care in Diabetesâ 2024 is available online and is published as a supplement to the January 2024 issue of Diabetes Care. - <https://diabetes.org/newsroom/press-releases/american-diabetes-association-releases-standards-care-diabetes-2024>
- Volume 47 Issue Supplement_1
- Diabetes Care - Standards of Care in Diabetesâ 2024 • 1. Improving Care and Promoting Health in Populations: Standards of Care in Diabetesâ 2024 • 2. Diagnosis and Classification ... - https://diabetesjournals.org/care/issue/47/Supplement_1
- Trends in First-Line Glucose-Lowering Drug Use in Adults With Type ... - Among adult patients with type 2 diabetes, metformin was by far the most frequent first-line treatment. While the use of SGLT-2i and GLP-1RA was ... - <https://diabetesjournals.org/care/article/44/8/1774/138868/Trends-in-First-Line-Glucose-Lowering-Drug-Use-in>

Care Guidance Summary for PT-4AE48A

- 9. Pharmacologic Approaches to Glycemic Treatment: Standards of ... - ... medication that historically has been the first-line treatment for type 2 diabetes (93,94). Metformin is effective and safe, is inexpensive ... - https://diabetesjournals.org/care/article/49/Supplement_1/S183/163934/9-Pharmacologic-Approaches-to-Glycemic
- 9. Pharmacologic Approaches to Glycemic Treatment: Standards of ... - Compared with sulfonylureas, metformin as first-line therapy has beneficial effects on A1C, is weight neutral, does not cause hypoglycemia, and ... - https://diabetesjournals.org/care/article/47/Supplement_1/S158/153955/9-Pharmacologic-Approaches-to-Glycemic

Next Steps

- Random glucose 130 mg/dL is mildly elevated and worth monitoring over time.
- Maintain the current glucose-friendly eating pattern to preserve HbA1c 5.2% and fasting glucose 90 mg/dL. Focus on eating a balanced diet with plenty of vegetables, whole grains, lean proteins, and healthy fats to help maintain stable blood sugar levels.
- Primary option for clinician review: Vitamin D (Cholecalciferol). Safe for general metabolic and bone health support; no known contraindications or interactions in this context. Low risk of adverse effects and may provide secondary benefits for metabolic health.
- Repeat HbA1c at the usual clinician-directed interval to confirm stability from the current 5.2% baseline.
- Monitor fasting or home glucose trends against the current 90 mg/dL baseline.